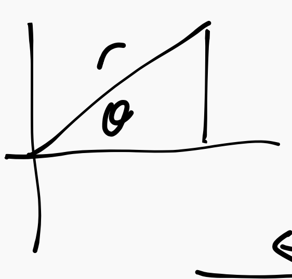
8.1

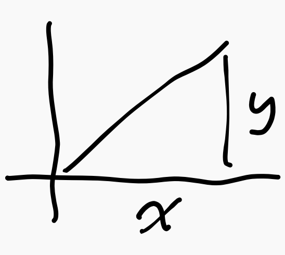
rect $\langle x,y\rangle$ polar (r,θ)

$p \rightarrow r$



$x = r\cos\theta$

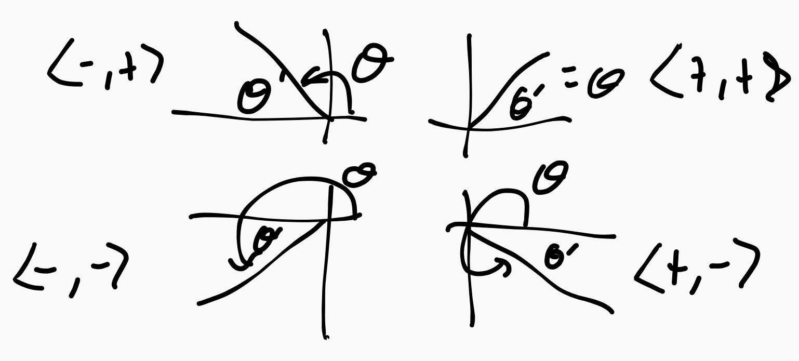
$y = r\sin\theta$



$r \rightarrow p$

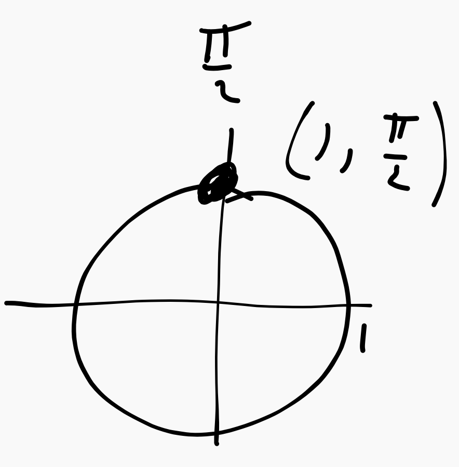
$r=\sqrt{x^2+y^2}$

$\theta' = \tan^{-1}\left|\frac{y}{x}\right|$

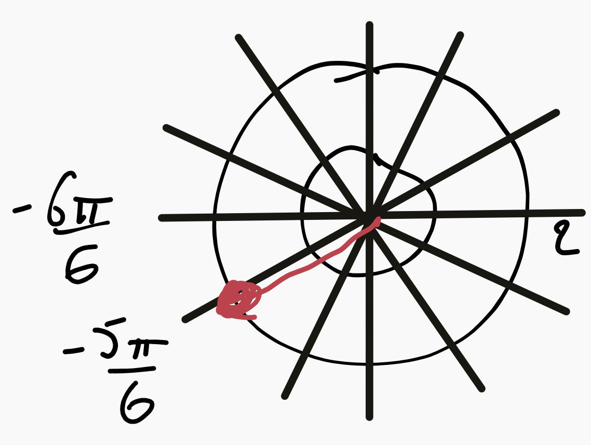


2) $(1, \frac{\pi}{2})$

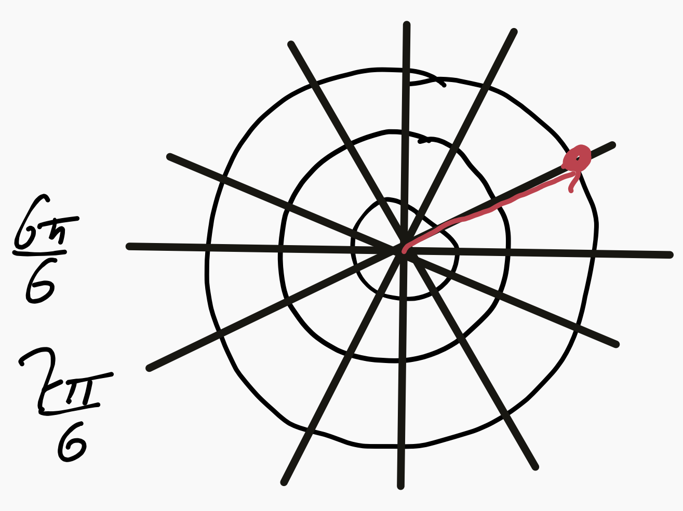
2nd ↗ ↑ 1st



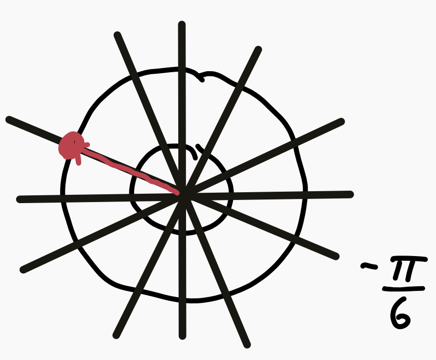
3) $(2, -\frac{5\pi}{6})$



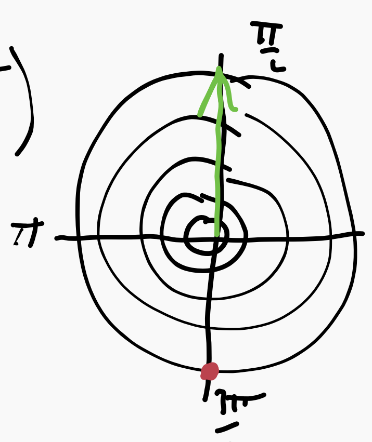
4) $(-3, \frac{7\pi}{6})$



5) $(-2, -\frac{\pi}{6})$



6) $(5, \frac{3\pi}{2})$



(r,θ)

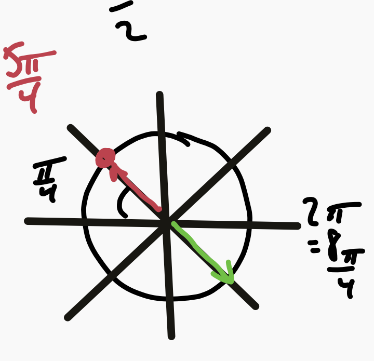
$r<0$	$r>0$
$(-5, \frac{\pi}{2})$	$(5, -\frac{\pi}{2})$
$(-5, -\frac{3\pi}{2})$	$(5, \frac{7\pi}{2})$

$\frac{3\pi}{2} + 2\pi$

$= \frac{3\pi}{2} + \frac{4\pi}{2} = \frac{7\pi}{2}$

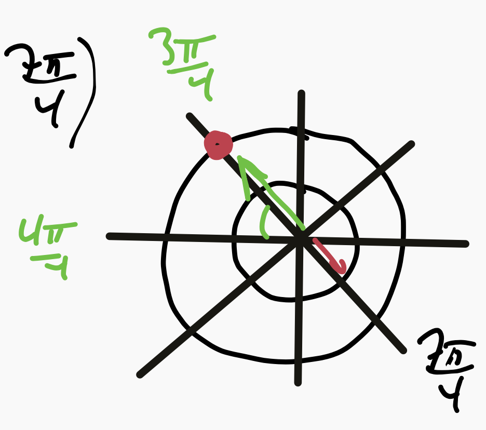
7) $(1, -\frac{5\pi}{4})$

$\frac{4\pi}{4} = -\frac{4\pi}{4}$



$r<0$	$r>0$
$(-1, -\frac{\pi}{4})$	$(1, \frac{3\pi}{4})$
$(-1, \frac{7\pi}{4})$	

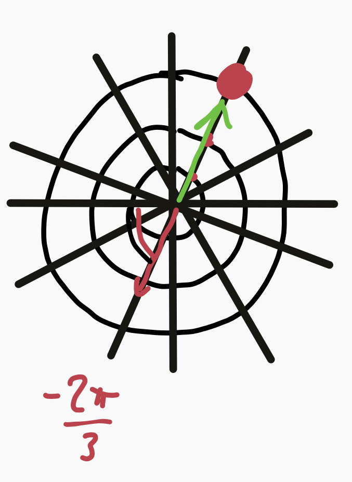
8) $(-2, \frac{7\pi}{4})$



$r<0$	$r>0$
$(-2, -\frac{\pi}{4})$	$(2, \frac{3\pi}{4})$

9) $(-3, -\frac{2\pi}{3})$

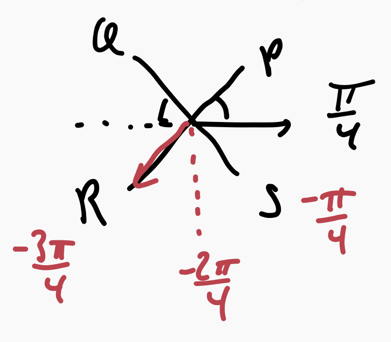
$\frac{3\pi}{3} = -\frac{3\pi}{3}$



$r<0$	$r>0$
$(-3, \frac{4\pi}{3})$	$(3, \frac{\pi}{3})$

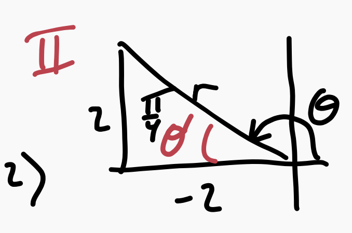
8.1.18) R

$(4, -\frac{3\pi}{4})$



8.1.25)

$\langle -2, 2\rangle$



$(\sqrt{8}, \frac{3\pi}{4})$

$r = \sqrt{(-2)^2+2^2}$

$= \sqrt{4+4}$

$= \sqrt{8}$

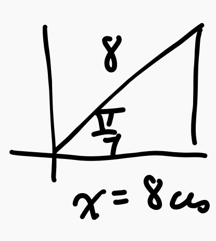
$\theta' = \tan^{-1}\left|\frac{2}{-2}\right| = \tan^{-1}\left|\frac{1}{1}\right| = \frac{\pi}{4}$

$\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

10) $(8, \frac{\pi}{4}) \rightarrow \langle x, y\rangle$

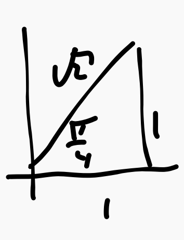
$\langle 8\frac{1}{\sqrt{2}}, 8\frac{1}{\sqrt{2}}\rangle$

$= \langle \frac{8}{\sqrt{2}}, \frac{8}{\sqrt{2}}\rangle$



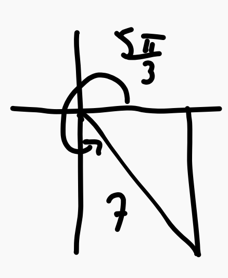
$y = 8\sin\frac{\pi}{4}$

$x = 8\cos\frac{\pi}{4}$



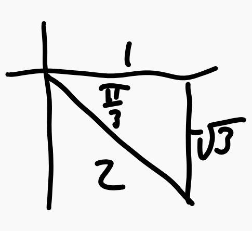
11) $(7, \frac{5\pi}{3}) \rightarrow \langle x, y\rangle$

$\frac{3\pi}{3} + 2\pi = \frac{6\pi}{3}$



$x = 7\cos\frac{7\pi}{3}$

$y = 7\sin\frac{7\pi}{3}$



$\langle 7(\frac{1}{2}), 7(\frac{-\sqrt{3}}{2})\rangle = \langle \frac{7}{2}, \frac{-7\sqrt{3}}{2}\rangle$

12) $(17, \frac{25\pi}{6}) \rightarrow \langle x, y\rangle$

$+\frac{12\pi}{6}$

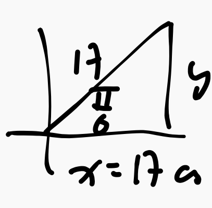
$\frac{24\pi}{6}$

$\frac{25\pi}{6} - \frac{24\pi}{6}$

$= \frac{\pi}{6}$

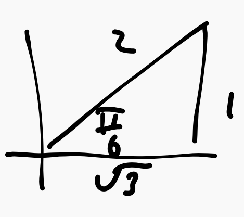
↑

$(17, \frac{\pi}{6})$



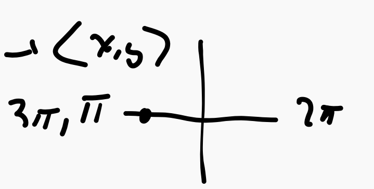
$y = 17\sin\frac{\pi}{6}$

$x = 17\cos\frac{\pi}{6}$



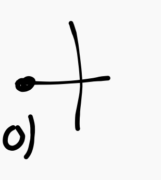
$\langle 17\frac{\sqrt{3}}{2}, 17\frac{1}{2}\rangle = \langle \frac{7\sqrt{3}}{2}, \frac{17}{2}\rangle$

13) $(3, 3\pi) \rightarrow \langle x, y\rangle$



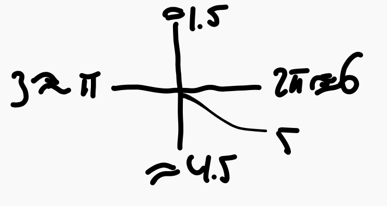
$x = 3\cos\pi$

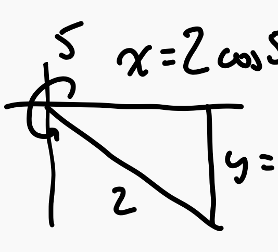
$y = 3\sin\pi$



$\langle -3, 0\rangle$

14) $(2, 5) \rightarrow \langle x, y\rangle$





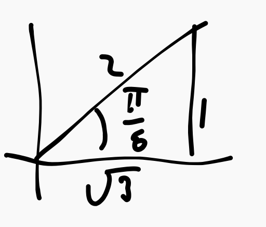
$x = 2\cos 5$

$y = 2\sin 5$

$\langle 2\cos 5, 2\sin 5\rangle$

15) $\langle \sqrt{3}, 1\rangle$

$\rightarrow (r, \theta)$



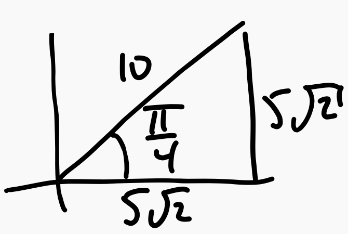
$(2, \frac{\pi}{6})$

16) $\langle 5\sqrt{2}, 5\sqrt{2}\rangle$

$\rightarrow (r, \theta)$

$(10, \frac{\pi}{4})$

a)



$r = \sqrt{(5\sqrt{2})^2 + (5\sqrt{2})^2}$

$= \sqrt{25(2) + 25(2)}$

$= \sqrt{100} = 10$

b) $\langle 5\sqrt{2}, 5\sqrt{2}\rangle$

$= 5\sqrt{2}\langle 1, 1\rangle$

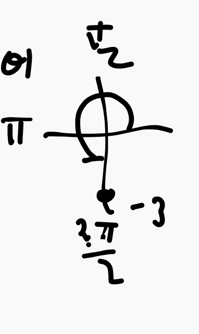
$= (5\sqrt{2}\cdot\sqrt{2}, \frac{\pi}{4})$

$= (10, \frac{\pi}{4})$

17) $\langle 0, -3\rangle \rightarrow (r, \theta)$

$r > 0$

$0 \le \theta < 2\pi$



$r = \sqrt{0^2 + (-3)^2} = 3$

$\theta' = \tan^{-1}\left|\frac{-3}{0}\right|$

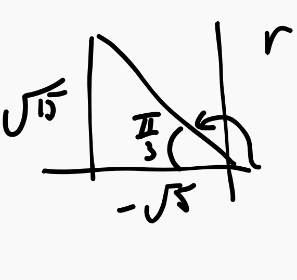
$\theta = \frac{3\pi}{2}$

$(3, \frac{3\pi}{2})$

18) $\langle -\sqrt{5}, \sqrt{15}\rangle \rightarrow (r, \theta)$

$\sqrt{3\cdot 5} = \sqrt{3}\sqrt{5}$

a)



$r = \sqrt{(-\sqrt{5})^2 + (\sqrt{15})^2}$

$= \sqrt{5 + 15}$

$= \sqrt{20}$

$\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

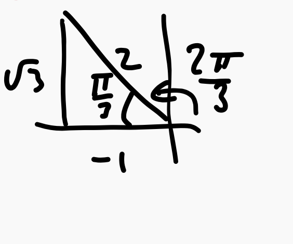
$(\sqrt{20}, \frac{2\pi}{3}) = (\sqrt{4}\sqrt{5}, \frac{2\pi}{3})$

$= (2\sqrt{5}, \frac{2\pi}{3})$

$\theta' = \tan^{-1}\left|\frac{\sqrt{15}}{-\sqrt{5}}\right| = \tan^{-1}\frac{\sqrt{15}}{\sqrt{5}}$

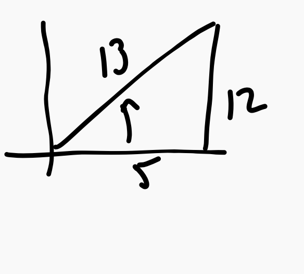
$= \tan^{-1}\frac{\sqrt{3}\cancel{\sqrt{5}}}{\cancel{\sqrt{5}}} = \tan^{-1}\frac{\sqrt{3}}{1}$

b) $\langle -\sqrt{5}, \sqrt{5}\sqrt{3}\rangle = \sqrt{5}\langle -1, \sqrt{3}\rangle$



$(\sqrt{5}\,2, \frac{2\pi}{3}) = (2\sqrt{5}, \frac{2\pi}{3})$

19) $\langle 5, 12\rangle \rightarrow (r, \theta)$

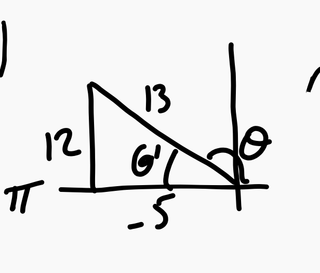


$r = 13$

$\theta' = \theta = \tan^{-1}\frac{12}{5}$

$(13, \tan^{-1}\frac{12}{5})$

19') $\langle -5, 12\rangle \rightarrow (r, \theta)$



$r = 13$

$\theta' = \tan^{-1}\left|\frac{12}{-5}\right| = \tan^{-1}\frac{12}{5}$

$\theta = \pi - \tan^{-1}\frac{12}{5}$

$(13, \pi - \tan^{-1}\frac{12}{5})$